Lesson plan · for the teacher

Standard deviation and outliers

Data science · Level 2 · Lesson 2

Time	Materials
40 min	Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Read a standard deviation.
- Flag outliers with the IQR rule.
- Check outliers before acting.

Activity: Spread detectives

1. Warm-up (5 min): Show two classes with a mean of 12, one tight and one spread out. Ask what a teacher would do differently.
2. Teach (8 min): Show distances from the mean as arrows; then find quartiles and draw the fences.
3. Find (15 min): Groups get a list of journey times with two odd values. They work out the fences and flag outliers.
4. Investigate (7 min): For each outlier, groups decide what could explain it and how to check.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- When does a large SD matter to a teacher?
- Why not delete outliers?
- Which outlier surprised you?

Exit ticket

- Q1 is 20 and Q3 is 30. Write the two fences.

Try it at school tomorrow
Take last term's marks without names, find the mean and the standard deviation, and look for outliers. Discuss what each outlier might mean.

Practice questions · for you

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1 **Class C has a standard deviation of 2, class D of 6. Which has more mixed marks?**

◯ Class D ◯ Class C ◯ They are the same

2 **An outlier should always be deleted so it doesn't spoil the average.**

◯ True ◯ False

3 **Print every mark that is more than two standard deviations above the mean, as a whole number.**

```
from ulab import numpy as np
marks = np.array([11, 12, 12, 13, 12, 30])
m = np.mean(marks)
s = np.std(marks)
print(m)
```

Write your program:

4 **Class A's marks are 11, 12, 12, 13. Class B's are 5, 9, 15, 19. Both have a mean of 12. Which has the larger standard deviation?**

◯ Class B ◯ Class A ◯ They are equal

5 **Complete the rule.**

Q1 is 10 and Q3 is 14, so the IQR is _____ . A mark above _____ is flagged as an outlier.

Word bank: 18 · 20 · 4 · 6

✎ **Think of a pupil whose numbers stand out. What would the standard deviation have hidden, and what does the outlier tell you?**

Check your answers

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1 **Class D**

A larger standard deviation means marks lie further from the mean.

2 **False**

It may be a real and important case. Check it first, fix real typos, and say what you did.

3

```
from ulab import numpy as np
marks = np.array([11, 12, 12, 13, 12, 30])
m = np.mean(marks)
s = np.std(marks)
for x in marks:
    if x > m + 2 * s:
        print(int(x))
```

The mean is 15 and the SD about 6.7, so the line is at about 28.5.

4 **Class B**

Class B's marks are much further from 12, so its standard deviation is larger.

5 **Q1 is 10 and Q3 is 14, so the IQR is 4. A mark above 20 is flagged as an outlier.**

IQR = 14 − 10 = 4, and the upper fence is 14 + 1.5 × 4 = 20.